

Investigating the Role of HbA1c Level in Predicting Diabetes Using Different Machine Learning Models

**CBIO313: Machine Learning & Data Mining
Project Report**

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1. Abstract

This project investigates the predictive power of the HbA1c level in diagnosing diabetes using different machine learning models on a large-scale health dataset of 100,000 individuals. The data includes demographic and clinical variables such as age, gender, BMI, blood glucose, smoking history, and hypertension status. A particular focus was placed on evaluating how well HbA1c levels indicate diabetes presence. Multiple classification models were trained and compared, including Random Forest, XGBoost, Gradient Boosting, AdaBoost, Logistic Regression, K-Nearest Neighbors, and others, using performance metrics such as accuracy, precision, recall, and F1 score. Ensemble methods — specifically Gradient Boosting, AdaBoost, and XGBoost — achieved the highest F1 scores of 0.81, offering the best balance between precision and recall. Results confirm the strong diagnostic relevance of HbA1c, reinforcing its clinical utility for early diabetes detection when combined with other health features. A web application (DiabetesIQ) was also deployed to demonstrate the end-to-end machine learning pipeline.

2. Introduction

Diabetes is a chronic metabolic disorder that affects millions of individuals globally and poses significant public health challenges. It is associated with severe long-term complications such as cardiovascular diseases, kidney failure, nerve damage, and vision loss if left undiagnosed or untreated. Therefore, early and accurate diagnosis is critical to managing and mitigating its progression. In clinical practice, Hemoglobin A1c (HbA1c) has emerged as a standard biomarker for the diagnosis and monitoring of diabetes. HbA1c reflects average blood glucose levels over a period of two to three months, offering a more stable measure compared to daily glucose readings.

Despite its wide acceptance, questions remain about the independent predictive power of HbA1c when used in isolation versus when combined with other demographic and clinical features. The availability of large-scale structured health data opens new opportunities to explore these relationships using machine learning techniques, which are capable of uncovering complex nonlinear interactions among variables.

This study aims to assess the diagnostic value of HbA1c using a variety of supervised classification models, ranging from simple linear classifiers to sophisticated ensemble methods. By incorporating additional features such as age, gender, BMI, blood glucose levels, smoking history, hypertension, and heart disease, we aim to build predictive models that can assist healthcare professionals in making data-driven decisions. The key objective is to compare the performance of these models and identify the one that provides the most reliable predictions in diagnosing diabetes.

3. Methodology

3.1 Dataset

The dataset comprises health records of 100,000 individuals (diabetes_dataset.csv) with the following features:

Feature	Description
age	Patient age
gender	Male / Female
bmi	Body Mass Index
hbA1c_level	Glycated hemoglobin — key predictor
blood_glucose_level	Blood glucose level
smoking_history	Smoking history category
hypertension	Hypertension status (0/1)
heart_disease	Heart disease status (0/1)
diabetes	Target variable (0 = No, 1 = Yes)

The dataset is highly imbalanced, with approximately 91% non-diabetic and 9% diabetic cases.

3.2 Data Preprocessing

- Verified no missing values or duplicate entries.
- Dropped the location column (no predictive value).
- Encoded categorical variables: gender (Male=0, Female=1) and smoking_history (No Info=0, never=1, former=2, current=3, not current=4, ever=5).
- Applied StandardScaler to: age, bmi, hbA1c_level, blood_glucose_level.
- Split data: 80% training / 20% testing (random_state=42).

3.3 Exploratory Data Analysis (EDA)

EDA was performed to uncover trends and relationships within the data. The following figures show the key visualizations generated.

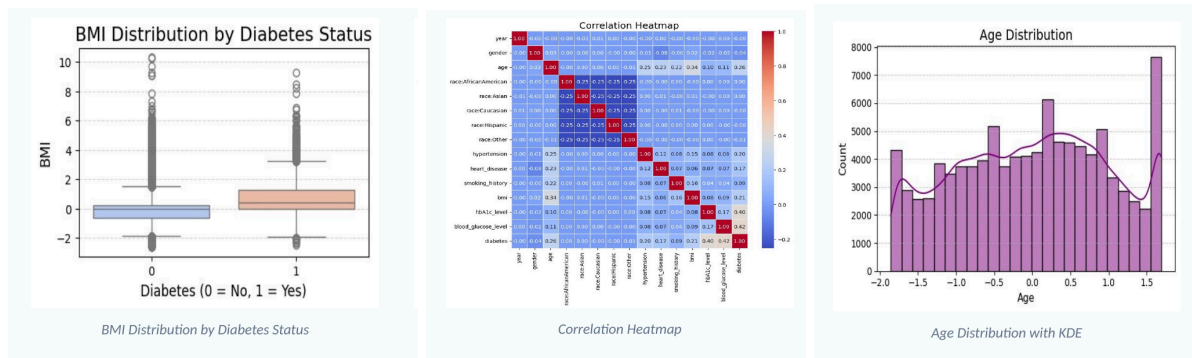


Fig. 1 — BMI Distribution by Diabetes Status, Correlation Heatmap, and Age Distribution with KDE

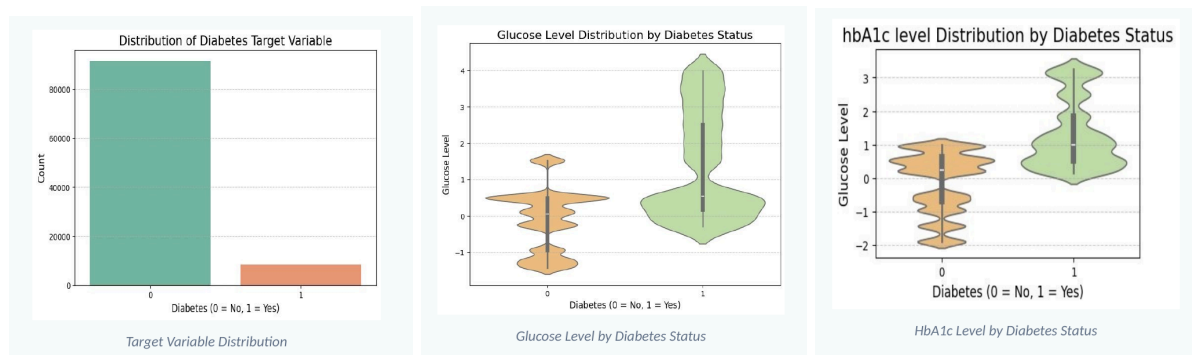


Fig. 2 — Target Variable Distribution, Glucose Level by Diabetes Status, and HbA1c Level by Diabetes Status

The violin plots confirm that both HbA1c level and blood glucose are significantly higher in diabetic patients, establishing them as the strongest predictors. The correlation heatmap reinforces this finding. The target distribution chart confirms a significant class imbalance (~91% non-diabetic).

3.4 Feature Engineering & Selection

A new binary feature `high_glucose` was engineered: `high_glucose = 1` if `blood_glucose_level > 140` (high risk), else 0. Feature selection was performed using an embedded method — Random Forest feature importances — to identify the top 5 most important features for final model training.

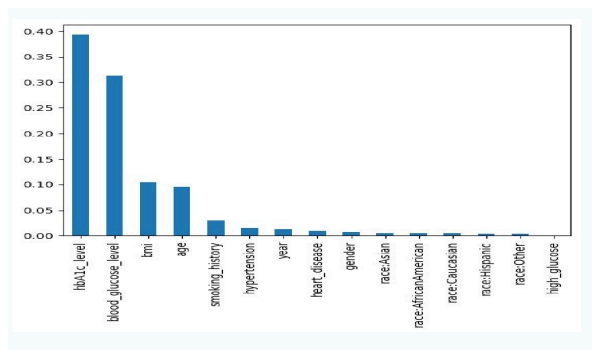


Fig. 3 — Random Forest Feature Importances (top 5 selected for final models)

The top 5 features were: hbA1c_level (~ 0.39), blood_glucose_level (~ 0.31), bmi (~ 0.10), age (~ 0.09), and smoking_history (~ 0.03). Logistic regression linearity check plots were also generated to verify monotonic relationships between each feature and the diabetes target.

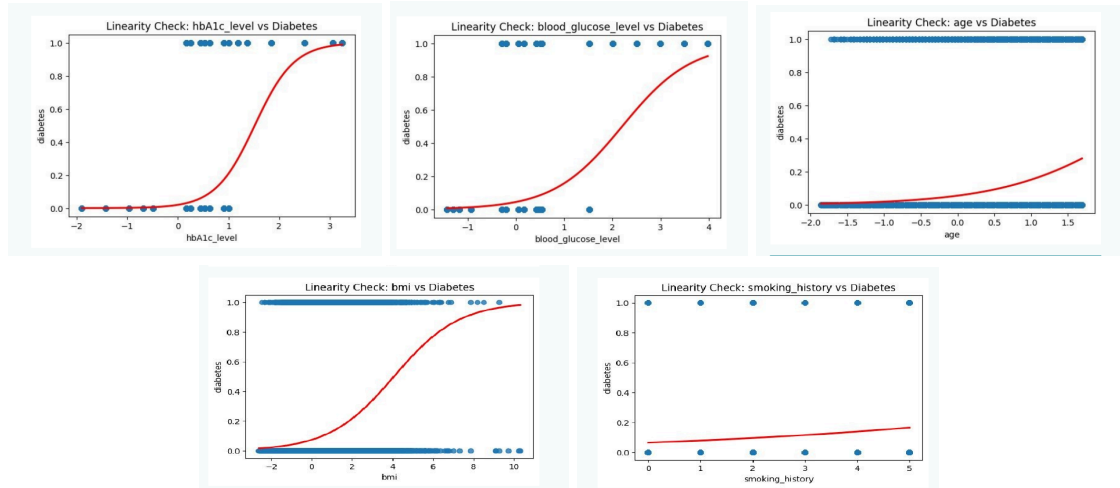


Fig. 4 — Linearity Checks: Logistic regression trend plots confirming monotonic feature-target relationships

3.5 Models Trained

Thirteen supervised classification models were trained and evaluated: Logistic Regression, KNN, Decision Tree, Naive Bayes, SVM, Random Forest, AdaBoost, Gradient Boosting, XGBoost, Bagging, Hard Voting, Soft Voting, and Stacking Classifiers. All models were evaluated using: Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC Curve with AUC Score.

4. Results & Discussion

The results from all 13 trained models are summarized in Table 1, followed by detailed discussion of each model with their confusion matrices and ROC curves.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.97	0.97	0.68	0.80
XGBoost	0.97	0.96	0.69	0.81
Gradient Boosting	0.97	0.99	0.68	0.81
AdaBoost	0.97	0.97	0.69	0.81
Bagging	0.97	0.94	0.70	0.80
Stacking	0.97	0.91	0.67	0.77
Logistic Regression	0.96	0.87	0.62	0.73
KNN	0.96	0.90	0.57	0.70
Hard Voting	0.9652	0.93	0.64	0.76
Soft Voting	0.9649	0.89	0.67	0.77

Model	Accuracy	Precision	Recall	F1 Score
Decision Tree	0.95	0.70	0.73	0.72
Naive Bayes	0.91	0.46	0.66	0.54
SVM	0.91	0.00	0.00	0.00

Table 1. Summary of ML model performance metrics across all 13 models

4.1 Random Forest

Random Forest achieved an accuracy of 0.97, precision of 0.97, recall of 0.68, and an F1 score of 0.80. The confusion matrix showed 18,264 true negatives and 1,154 true positives with only 27 false positives (AUC = 0.96). It consistently handled feature importance and complex interactions effectively.

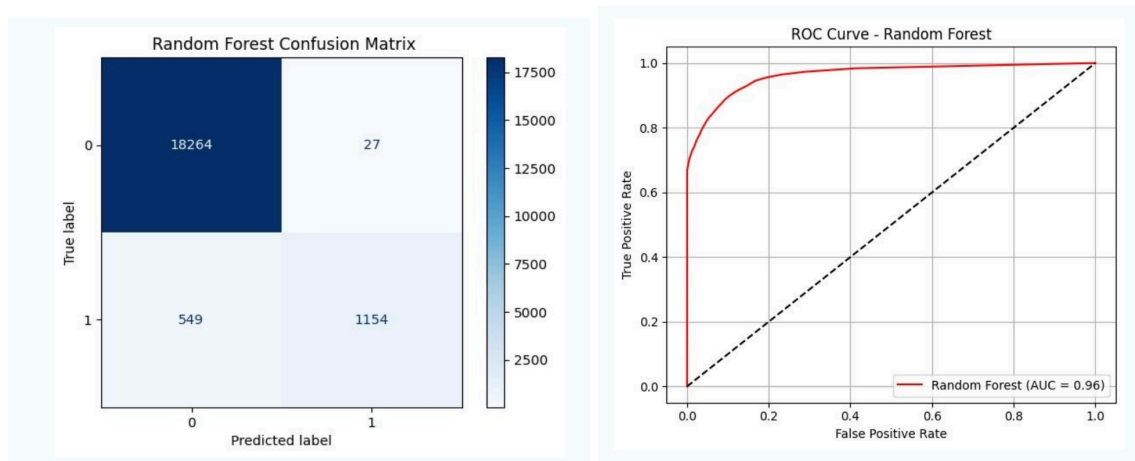


Fig. 5 — Random Forest: Confusion Matrix and ROC Curve (AUC = 0.96)

4.2 K-Nearest Neighbors (KNN)

KNN showed reasonable results with 0.96 accuracy, 0.90 precision, 0.57 recall, and 0.70 F1 score. The confusion matrix showed 18,182 true negatives and 971 true positives (AUC = 0.89). While precision was high, the relatively low recall indicates challenges in detecting all diabetic cases.

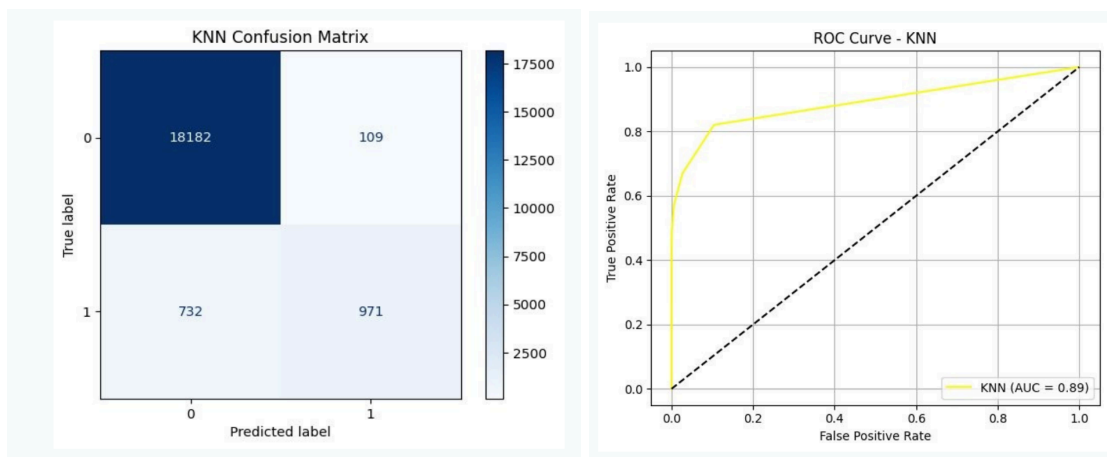


Fig. 6 — KNN: Confusion Matrix and ROC Curve ($AUC = 0.89$)

4.3 Logistic Regression

Logistic Regression achieved 0.96 accuracy, 0.87 precision, 0.62 recall, and 0.73 F1 score. The confusion matrix showed 18,138 true negatives and 1,065 true positives ($AUC = 0.96$). Its strength lies in interpretability, but it struggled to capture complex nonlinear patterns compared to ensemble methods.

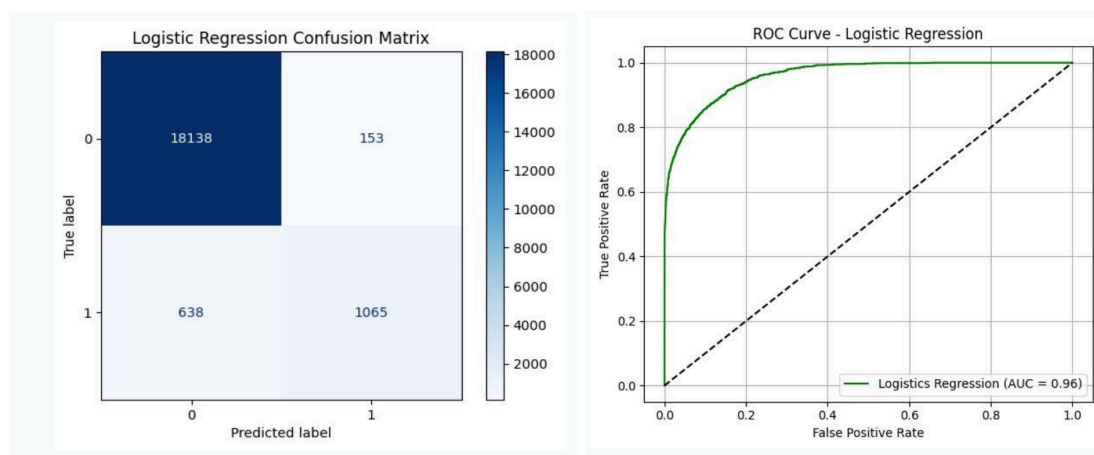


Fig. 7 — Logistic Regression: Confusion Matrix and ROC Curve ($AUC = 0.96$)

4.4 Decision Tree

Decision Tree delivered 0.95 accuracy, 0.70 precision, 0.73 recall, and 0.72 F1 score ($AUC = 0.85$). With 17,753 true negatives and 1,242 true positives, it had the highest recall of all non-ensemble models, offering a clear and interpretable structure for identifying diabetes risk.

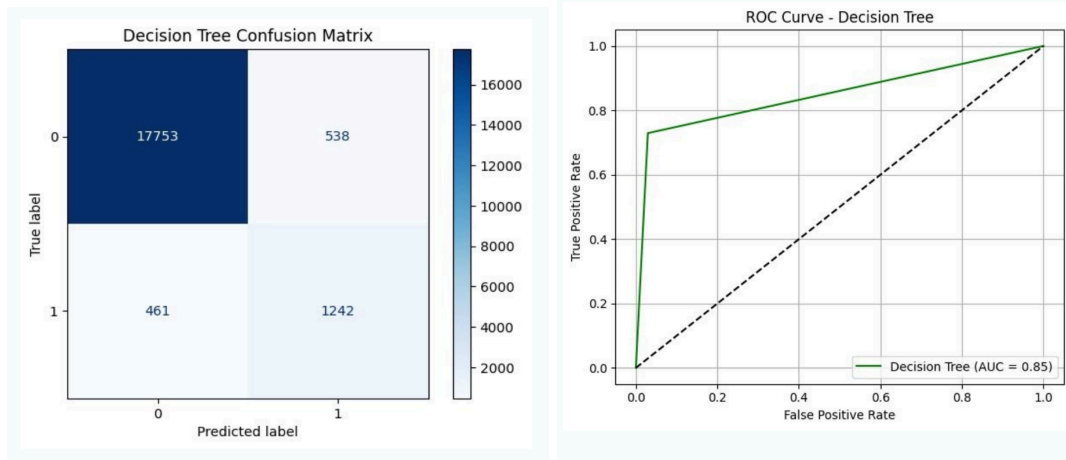


Fig. 8 — Decision Tree: Confusion Matrix and ROC Curve (AUC = 0.85)

4.5 AdaBoost

AdaBoost achieved 0.97 accuracy, 0.97 precision, 0.69 recall, and 0.81 F1 score (AUC = 0.97). Notably, the confusion matrix showed 0 false positives — the highest specificity of all models — making it extremely reliable at avoiding false alarms.

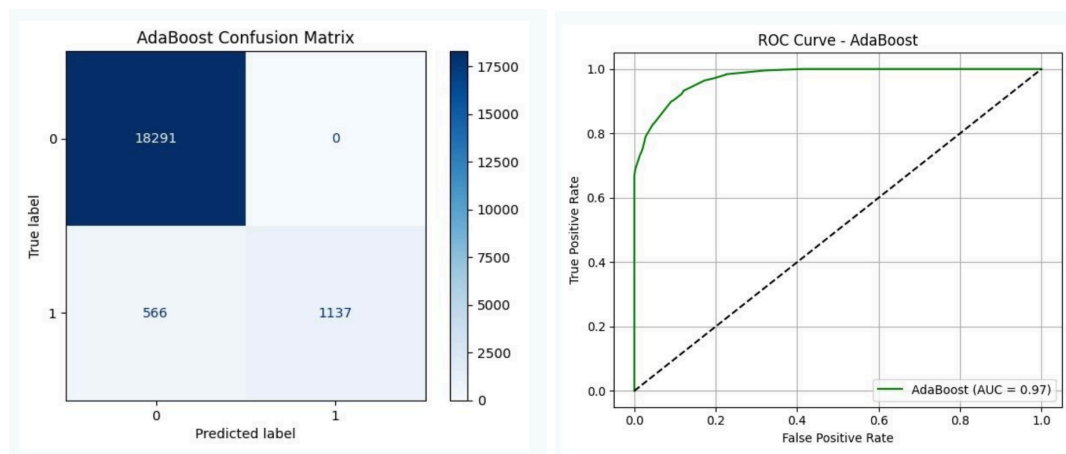


Fig. 9 — AdaBoost: Confusion Matrix and ROC Curve (AUC = 0.97)

4.6 Gradient Boosting

Gradient Boosting achieved 0.97 accuracy, a standout precision of 0.99, 0.68 recall, and 0.81 F1 score (AUC = 0.98). With only 14 false positives — the fewest of all models — it is extremely confident in positive predictions and tied for the highest F1 score.

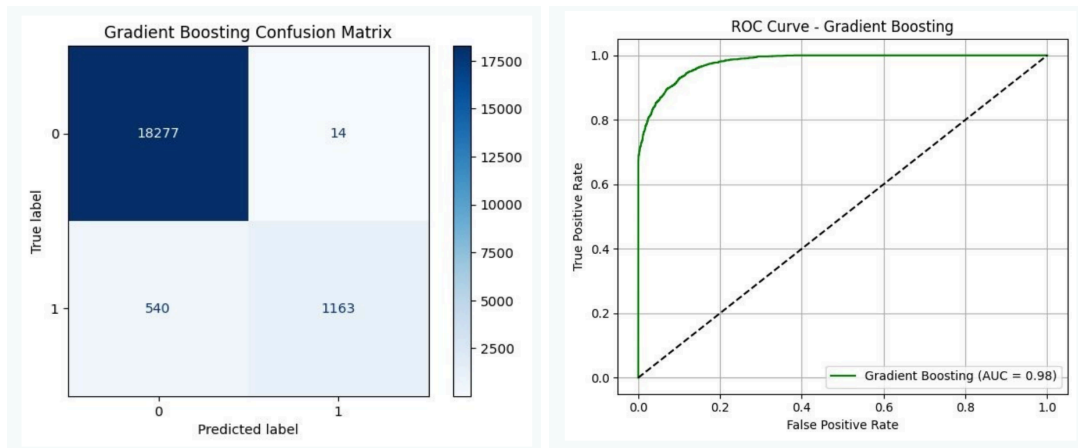


Fig. 10 — Gradient Boosting: Confusion Matrix and ROC Curve (AUC = 0.98)

4.7 XGBoost

XGBoost delivered 0.97 accuracy, 0.96 precision, 0.69 recall, and 0.81 F1 score (AUC = 0.98). With 18,247 true negatives and 1,180 true positives, it tied for the highest F1 score and is particularly suitable for scalable training on large datasets.

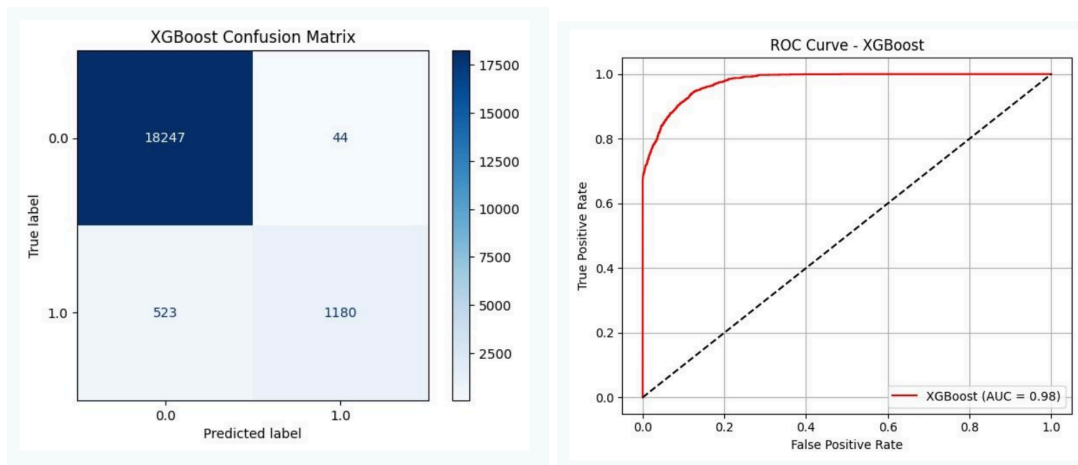


Fig. 11 — XGBoost: Confusion Matrix and ROC Curve (AUC = 0.98)

4.8 Stacking Classifier

Stacking Classifier achieved 0.97 accuracy, 0.91 precision, 0.67 recall, and 0.77 F1 score (AUC = 0.96). While strong overall, its F1 score was slightly lower than the top ensemble models, indicating a moderate tendency to miss some positive cases.

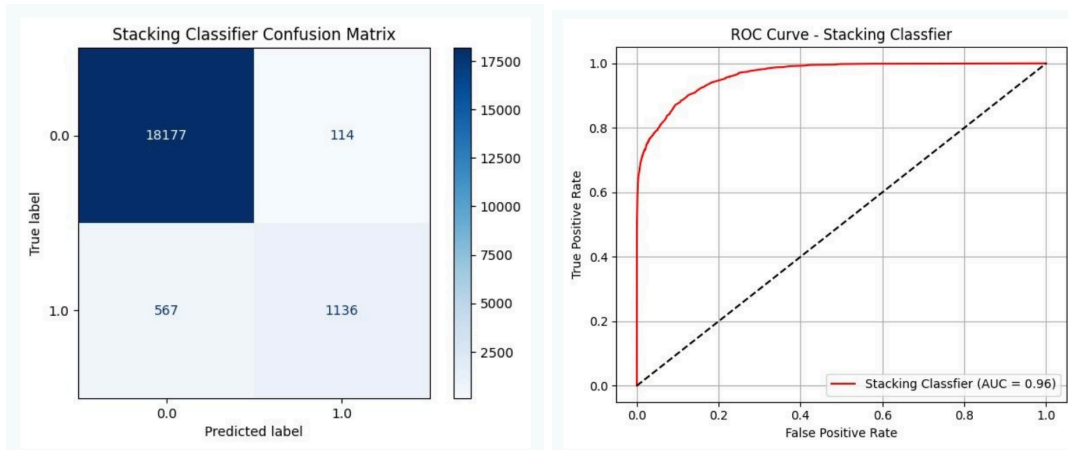


Fig. 12 — Stacking Classifier: Confusion Matrix and ROC Curve (AUC = 0.96)

4.9 Bagging Classifier

Bagging Classifier demonstrated robust performance with 0.97 accuracy, 0.94 precision, 0.70 recall, and 0.80 F1 score (AUC = 0.96). With 18,220 true negatives and 1,192 true positives, it offers strong stability and interpretability.

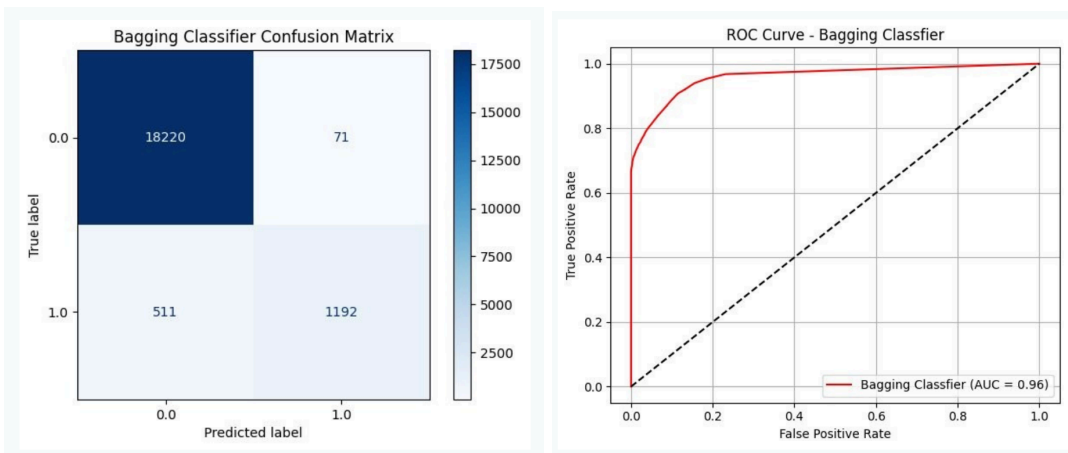


Fig. 13 — Bagging Classifier: Confusion Matrix and ROC Curve (AUC = 0.96)

4.10 Naive Bayes & SVM

Naive Bayes showed a noticeable drop in performance with 0.91 accuracy, 0.46 precision, 0.66 recall, and 0.54 F1 score (AUC = 0.92). Its assumption of feature independence limited its ability to handle complex clinical feature interactions.

Support Vector Machine (SVM) performed poorly despite 0.91 accuracy, with near-zero precision, recall, and F1 score. The confusion matrix revealed SVM predicted all instances as negative, completely failing to detect any diabetic case due to class imbalance sensitivity.

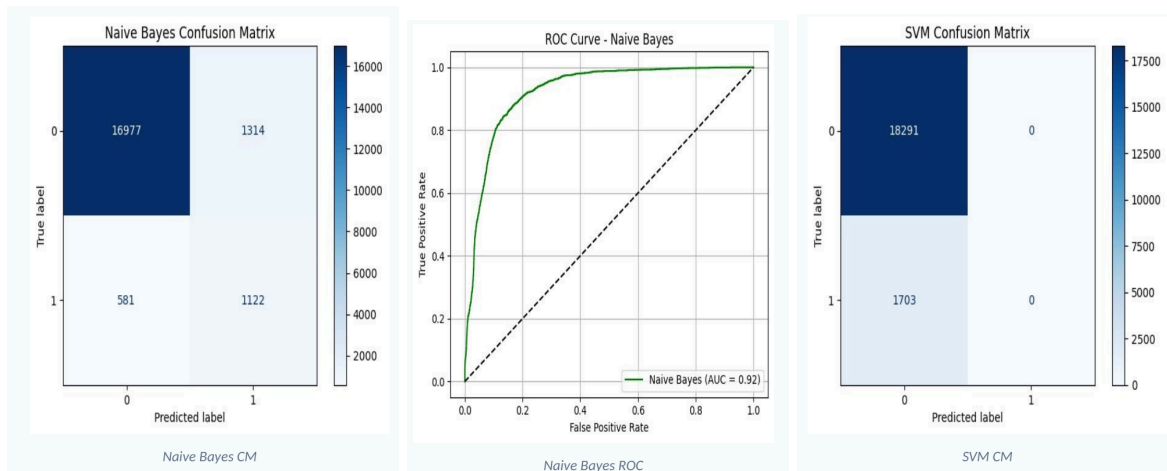


Fig. 14 — Naive Bayes (Confusion Matrix & ROC AUC = 0.92) and SVM (Confusion Matrix)

4.11 Hard Voting & Soft Voting

Hard Voting reported 0.9652 accuracy, 0.93 precision, 0.64 recall, and 0.76 F1 score. **Soft Voting** scored 0.9649 accuracy, 0.89 precision, 0.67 recall, and 0.77 F1 score. Soft Voting showed slightly more balanced behavior between false positives and false negatives.

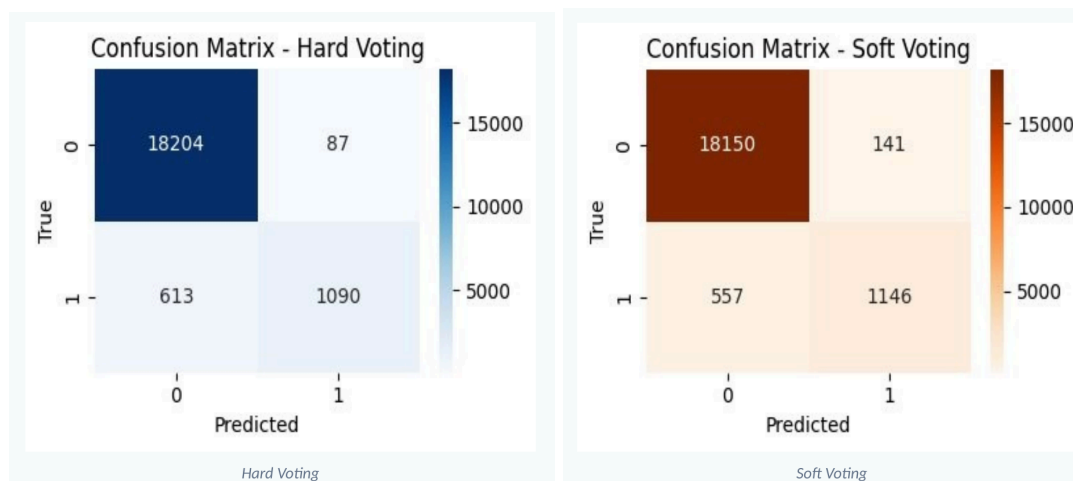


Fig. 15 — Hard Voting and Soft Voting: Confusion Matrices

4.12 Overall Performance Comparison

The grouped bar chart below visualizes accuracy, precision, recall, and F1 scores across all 13 models. Gradient Boosting, XGBoost, and AdaBoost cluster at the top with the highest F1 scores of 0.81. SVM and Naive Bayes fall significantly behind, especially in precision and F1 score.

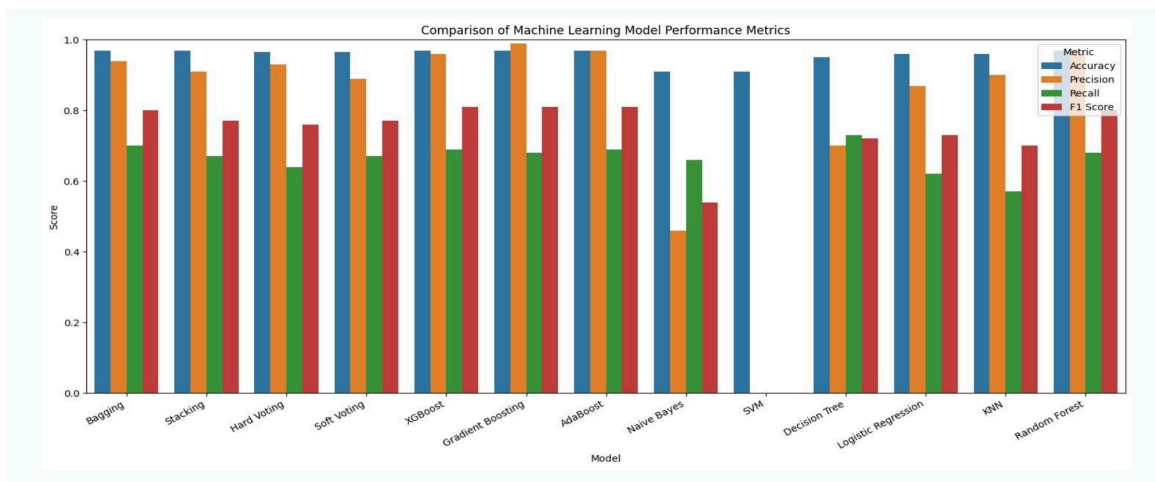


Fig. 16 — Comparison of all ML model performance metrics

5. Deployment

The best-performing model pipeline was saved using the joblib library (model.pkl) and deployed as an interactive web application named DiabetesIQ. The application was built using Streamlit and provides a user-friendly diabetes risk assessment interface.

The application accepts the following clinical inputs: Age (years), BMI (kg/m²), HbA1c Level (%), and Blood Glucose (mg/dL) — all via interactive sliders with clinical reference ranges. Upon clicking "Analyze Risk Profile", the model returns a prediction result along with probability scores for three classes: No Diabetes, Pre-Diabetes, and Diabetes.

The screenshot shows the DiabetesIQ Diabetes Risk Predictor interface. It features a dark background with a red header bar containing the logo and title. Below the title, there is a subtitle and a brief instruction. The main area contains four horizontal sliders for inputting clinical measurements: Age (years), BMI (kg/m²), HbA1c Level (%), and Blood Glucose (mg/dL). Each slider has a red dot indicating the current value and a red line showing the range. Below the sliders, there is a button labeled "Clinical reference ranges" and a large red button labeled "Analyze Risk Profile".

Fig. 17 — DiabetesIQ input interface showing clinical parameter sliders

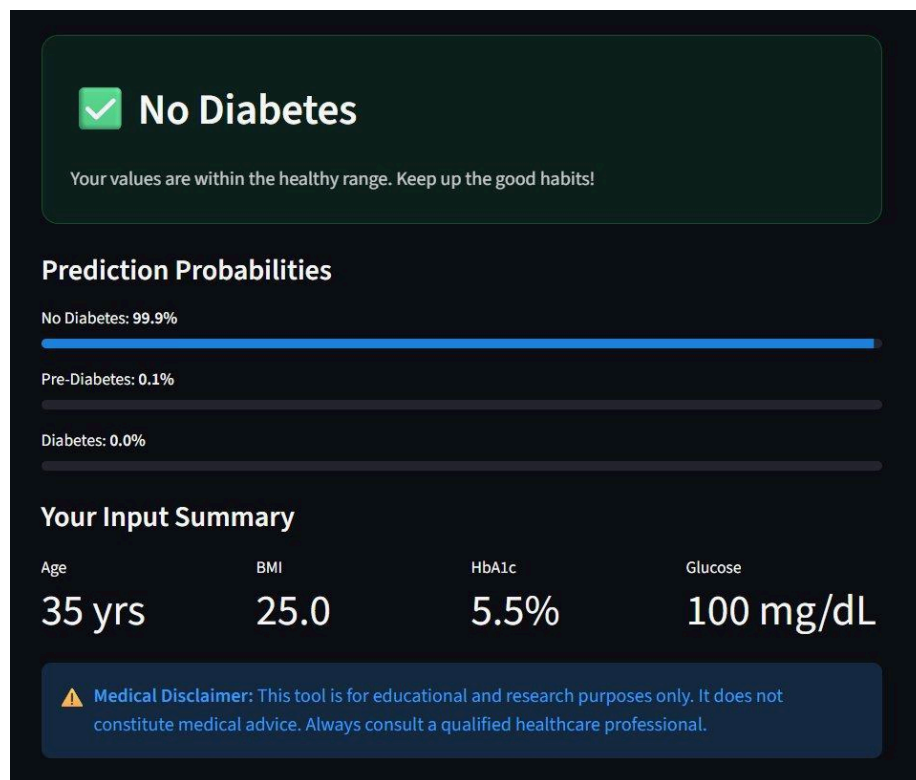


Fig. 18 — DiabetesIQ output: "No Diabetes" result with 99.9% confidence for a 35-year-old patient (BMI=25.0, HbA1c=5.5%, Glucose=100 mg/dL)

A medical disclaimer is displayed indicating that the tool is for educational and research purposes only. The application was tested locally and demonstrates a complete end-to-end ML pipeline from data ingestion and model training to real-time clinical risk assessment.

6. Conclusion

This study confirmed that HbA1c is a highly informative feature in machine learning models for predicting diabetes. The feature importance analysis clearly ranked hbA1c_level (~0.39) and blood_glucose_level (~0.31) as the two dominant predictors, together accounting for approximately 70% of the model's predictive power — consistent with their established clinical significance.

Among the 13 models evaluated, ensemble methods — particularly Gradient Boosting, AdaBoost, and XGBoost — achieved the highest F1 scores of 0.81 and AUC scores of 0.97–0.98. Random Forest and Bagging also performed strongly. In contrast, SVM completely failed to detect diabetic cases due to class imbalance, and Naive Bayes was limited by its independence assumption.

The project highlights the importance of evaluating models beyond accuracy alone, since most achieved ~0.97 accuracy due to class imbalance while F1 scores ranged from 0.00 to 0.81. The DiabetesIQ web application further demonstrates the practical value of machine learning in healthcare. Future directions include addressing class imbalance (SMOTE), hyperparameter tuning for SVM, external dataset validation, and multi-class classification for prediabetes stages.

References

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Diabetes Prediction Dataset — 100,000 patient records (diabetes_dataset.csv)
<https://www.kaggle.com/datasets/priyamchoksi/100000-diabetes-clinical-dataset/data>

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